6. Widal Test (Slide Test)

Aim

To carry out the widal test for the given blood sample and to determine the presence of antibodies against salmonella antigens.

Theory And Principle

Widal test is a serological test which is used for the diagnosis of enteric fever or typhoid fever. Typhoid or enteric fever is caused by a gram negative bacteria *Salmonella enterica* (*Salmonella* Typhi or *Salmonella* Paratyphi). *Salmonella* possess O antigen on their cell wall and H antigen on their flagella. On infection, these antigen stimulates the body to produce specific antibodies which are released in the blood. The Widal test is used to detect these specific antibodies in the serum sample of patients suffering from typhoid using antigen-antibody interactions. These specific antibodies can be detected in the patient's serum after 6 days of infection (fever).

Salmonella Typhi possesses O antigen on the cell wall and H antigen on flagella. *Salmonella* Paratyphi A and *S.* Paratyphi B also possess O antigen on their cell wall and but have AH and BH antigen on their flagella respectively.

Widal test is an agglutination test in which specific typhoid fever antibodies are detected by mixing the patient's serum with killed bacterial suspension of Salmonella carrying specific O, H, AH and BH antigens and observed for clumping ie. Antigen-antibody reaction. The main principle of Widal test is that if homologous antibody is present in patient's serum, it will react with respective antigen in the suspension and gives visible clumping on the test slide.

Requirements

Fresh serum

The complete kit containing five vials containing stained Salmonella antigen

- *S.* Typhi ⟶ O antigen
- *S.* Tyhhi ⟶ H antigen
- *S.* Paratyphi ⟶ AH antigen
- *S.* Paratyphi ⟶ BH antigen

Widal positive control

Widal test card or slide

v) Applicator stick

Procedure

- Widal test can be done in two ways-one is rapid test on slide and another is tube test in which result may be obtained after one night of incubation.

XII_Micro_Biology_Practical_Manual.indd 302 15-04-2019 17:13:15